

Title:

Hazardous Scream Detection and Alert System

Project Purpose:

The objective of this project is to develop an advanced scream detection system capable of identifying human screams in real time, particularly in situations where an individual may be in danger. Upon detection, the system will automatically trigger an alert, notifying emergency responders such as police and pre-set emergency contacts. This initiative aims to enhance safety by ensuring a rapid response in potentially life-threatening situations, leveraging machine learning and real-time monitoring.

Project Scope:

This project focuses on the development of a system that can efficiently differentiate between screams and background noise, ensuring accurate alerts while minimizing false positives.

The system will utilize continuous, real-time scream detection and alert mechanisms, ensuring rapid response times in emergency situations.

Key Components and Tasks:

1. A Scream Detection ML Model

- Utilize machine learning and audio signal processing techniques to accurately identify human screams.
- Train the model to differentiate screams from general background noises, such as traffic, music, or conversations.

2. Automated Alert System using Twilio

- Develop a mechanism to instantly send SMS alerts to emergency contacts and local authorities when a scream is detected.
- Ensure that the alerts contain relevant location details and other necessary information for a swift response.

3. Real-Time Monitoring & Deployment

- Design the system to operate 24/7, continuously monitoring audio input for distress signals.
- Optimize the system for both indoor and outdoor environments.

4. System Testing

- Conduct extensive testing in different acoustic environments to refine detection accuracy.
- Implement strategies to reduce false alarms while maintaining high sensitivity to actual distress signals.

Project Boundaries:

- The system will be software-based, utilizing existing microphone setups for audio input, without incorporating dedicated physical hardware.
- Its primary focus will be on detecting human screams that indicate distress, rather than general loud noises or sudden sounds.

Deliverables:

- ✓ A fully functional scream detection Algorithm capable of real-time analysis.
- ✓ An integrated emergency alert system that sends SMS notifications upon scream detection.
- ✓ Comprehensive documentation covering system design, implementation, testing, and performance evaluations.